

UGTS-KC 3.6.2 – SCLP

Swept-Cone Log-Polar Packing

Formal geometric-topological and kinematic delta over 3.6.1 BEA

Integrated source mechanisms

A finite cone defined by slant length and half-angle; an exact cone signed-distance relation; a certified translational sweep bound; two local spherical supports; a log-polar chart with metric, Jacobian, velocity and acceleration calculus; a deterministic one-bit perturbation interval; time-to-phase winding; a source half-turn bundle twist and a separate reflective Klein quotient; a missing-shackle constraint-release certificate; a bounded binary L-system; distinct contiguous and Morton 64-bit keys; and an audited memory/complexity boundary.

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Requester-supplied attribution; identifiers not independently verified.

Version 3.6.2 – generated 17 August 2026

The attached source is distilled without historical, dialect, renderer or physical-chaos escalation.

Abstract

The attached 13-page source collects a dense chain of motifs: a sweeping cone and sphere relations; analytical one-bit jitter; a log-encoded polar look-up space; lower-case ϕ as a hoop coordinate; angular “change of change” at a torque hinge; a missing shackle; a Klein-bottle-style wrap; time winding around an SDF-zero coil; bit shifts and binary bifurcation; a parametric L-system; instant state lookup; a 64-bit coordinate key; pointerless tree storage; and aggressive compression claims. The document also explicitly removes rasterization, ray marching and rendering from the intended core.

Version 3.6.2 converts those motifs into a finite referential substrate. It preserves the source terminology where technically meaningful, while separating overloaded symbols, coordinate effects, physical dynamics, topology, indexing and information loss. The resulting SCLP profile contains 50 atomic delta operators, 22 content-addressed definitions, a JSON Schema, two exact 64-bit key codecs, a dependency-free Python reference runtime, 70 new executable tests and the retained 83 tests from versions 3.6/3.6.1.

FORMAL DECISION

The authoritative object is not an image, mesh, ray-marched volume or frame loop. It is a typed state-and-relation package that can answer geometric, topological, kinematic, branch, key and event queries. The output returns to the ordinary UGTS sequence: support → compatibility → guard → verified event → transition → lineage.

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1 Source scope and extraction discipline

1.1 Registered source

The immediate source is a 13-page exported dialogue whose first prompt combines a sweeping cone, one-bit jitter, the cone's triangular side and circular base, a sphere, a torque hinge, two spherical "eyes", a log-polar LUT, a Klein-bottle motif, a binary tree and lower-case ϕ . Its later pages progressively remove the graphics interpretation and request a purely packed geometric-topological substrate, a formal grammar, instant state lookup, a 64-bit allocation and exact compression metrics.

SHA-256: 7e29c1c800d905268f35084a6a4c7d9c1cfed50c926c63a8ec2c79021e32ab63

Pages: 13

Raw source redistributed in ZIP: no

The relevant page groups are:

- pages 1–3: finite-cone/SDF language, one-bit jitter, spheres, log-polar coordinates, local radial step and angular acceleration;
- pages 3–5: time winding, chirality inversion, binary traversal and branch selection;
- pages 6–8: a branch-conditioned L-system and a proposed instantaneous movement vector;
- pages 9–11: explicit removal of rendering, a one-bit radix structure, a topological twist and a static typed state lookup;
- pages 11–13: the 20/18/14/12 allocation, Morton-style interleaving, pointerless storage and width/compression claims.

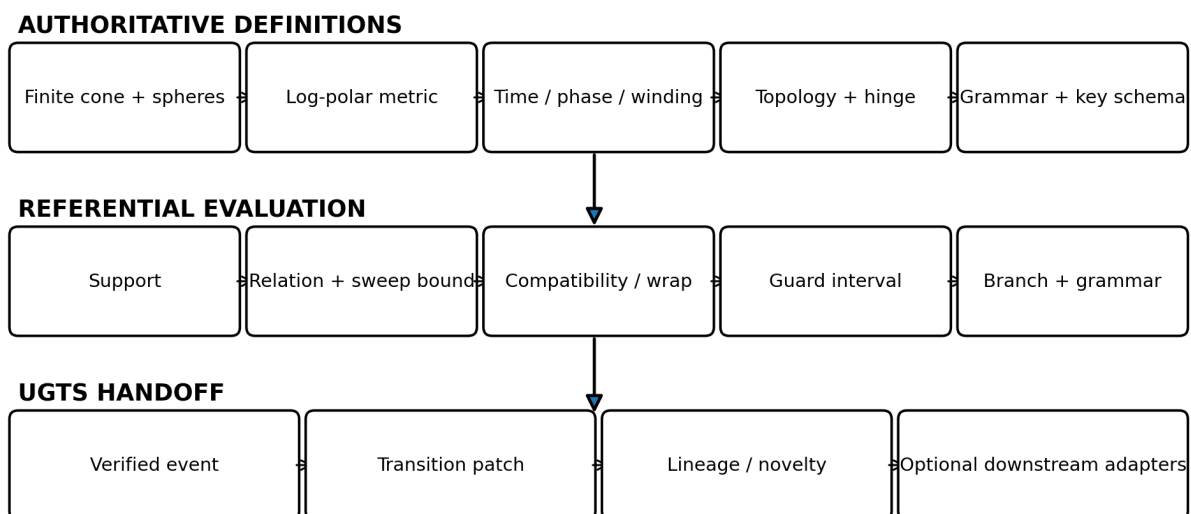
1.2 Admission rule

A source motif enters the 3.6.2 core only if it becomes one of the following: a typed coordinate, a finite relation, a support, a metric/Jacobian, a transition/gluings map, a branch rule, a finite grammar, a content-addressed definition, a finite key codec, a measurable error bound or an explicit claims-ledger entry.

EVIDENCE BOUNDARY

The source includes useful equations and also overstatements. Version 3.6.2 does not silently replace the source. It records both the source form and the normalized form. In particular, the source half-turn map is preserved as its own bundle profile; the actual orientation-reversing Klein profile is separate.

2 Canonical 3.6.2 architecture



No rasterization, ray marching or display authority in the 3.6.2 core

Figure 1: SCLP definitions and evaluation remain upstream of the canonical UGTS event handoff.

The typed state is expanded to

$$q = (x, t, \rho, \theta, \phi, \dot{\phi}, \ddot{\phi}, o, s, b, \omega_L, K, a, \mathcal{L}, u),$$

where x is position; t is linear time; (ρ, θ) is a local chart; $\phi \in \mathbb{S}^1$ is the hoop/hinge angle; $o \in \{-1, +1\}$ is orientation; s is a sheet; b is a branch bit; ω_L is a finite grammar-state word; K is a 64-bit address; a is a generative address; $\mathcal{L}$ is lineage; and u is uncertainty.

2.1 Symbol contract

The source overloads several symbols. SCLP imposes this non-negotiable contract:

Symbol	Typed use
T	cone slant length; never time, torque or a bit shift
t	continuous linear time
X	a declared modular tick or lookup-time coordinate
ϕ	periodic hinge/hoop angle in $\mathbb{S}^1$; not the golden ratio Φ
$\Delta\rho$	log-radius increment
Δt	time increment
ϵ_j	bounded jitter amplitude
ϵ_g	event guard margin
b	branch, predicate or jitter bit; never full state

3 Finite cone, circle and sphere relations

3.1 Cone parameterization

A side length alone does not determine a right circular cone. SCLP retains the source's side parameter T and adds the required half-angle α :

$$h = T \cos \alpha, \quad R = T \sin \alpha,$$

with $T > 0$ and $0 < \alpha < \pi/2$. The circular base is the disk of radius R in the plane $z = h$ relative to the apex/axis frame.

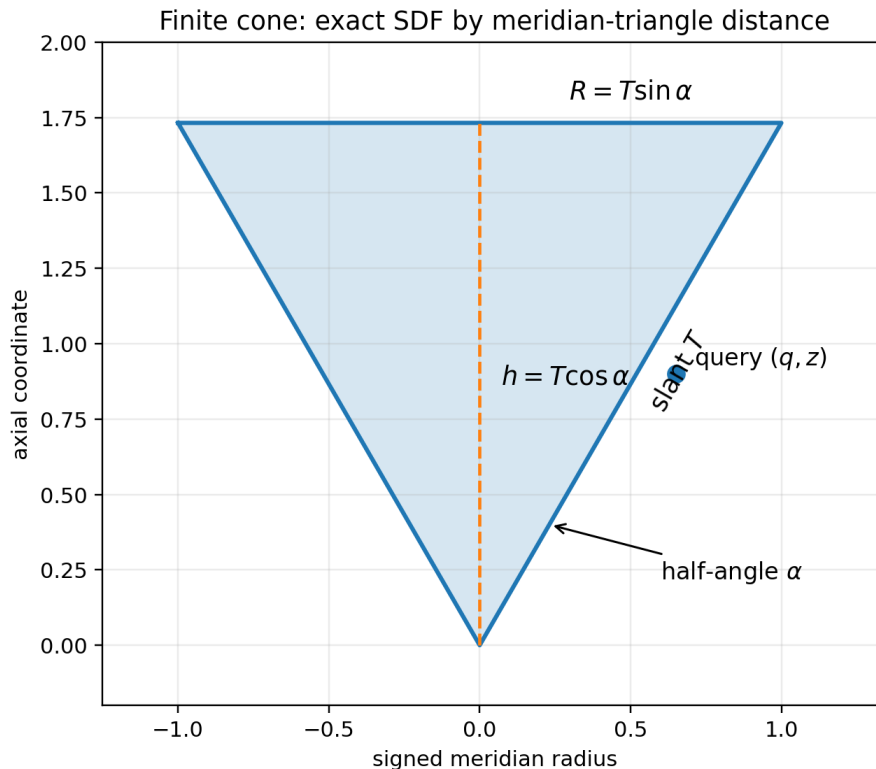


Figure 2: The triangular side view and circular solid are joined by rotational symmetry.

For apex c , unit axis a and query x , define

$$z = a \cdot (x - c), \quad q = \|(x - c) - za\|.$$

The exact Euclidean signed distance is the signed distance from (q, z) to the filled meridian triangle with vertices $(-R, h)$, (R, h) and $(0, 0)$. The sign is negative exactly when

$$0 \leq z \leq h, \quad q \leq z \tan \alpha.$$

3.2 Paired spherical support

The source's "two-sphered eye" is normalized as two local sphere relations

$$f_L(x) = \|x - c_L\| - R_L, \quad f_R(x) = \|x - c_R\| - R_R.$$

A profile chooses union, intersection or either-with-tag admission. The overlap flag is retained separately, so the two supports can function as a binocular pair, a dual sensor, two compatibility chambers or a local shared-domain test without a biological claim.

4 Sweeping cone as a certified relation family

Let C be the finite cone and let $s(u)$ be a finite translation path for $u \in [0, 1]$. The sweep relation is

$$F_{\text{sweep}}(x) = \inf_{u \in [0, 1]} d_C(x - s(u)).$$

For a linear segment of length L , $n \geq 2$ uniform samples and sample minimum m_n , signed distance under translation is 1-Lipschitz. Therefore

$$m_n - \frac{L}{2(n-1)} \leq F_{\text{sweep}}(x) \leq m_n.$$

DERIVATION / INVARIANT

The closest parameter u^* lies no farther than $1/[2(n-1)]$ from some sample. Translating the cone by a spatial distance δ changes its signed distance at a fixed query by at most δ . The interval follows. It becomes exact for a zero-length sweep and shrinks monotonically with sample density.

The reference instance uses $L = 0.75$, $n = 65$ and yields

$$-0.3024764633 \leq F_{\text{sweep}} \leq -0.2966170883,$$

so the query has a certified interior witness under a 0.001 guard band.

EVIDENCE BOUNDARY

This is not a universal closed-form SDF for every swept, rotated or grammar-generated cone. The exact primitive SDF and the sweep interval are different typed objects. Rotating sweeps require an additional angular-speed/bounding-radius error term or a specialized solver.

5 Log-polar metric and kinematic calculus

5.1 Coordinate chart and explicit core

The chart is

$$\rho = \ln(r/r_0), \quad \theta = \text{atan2}(y, x), \quad r = r_0 e^\rho.$$

The origin remains singular. Every profile therefore declares a core radius and returns a core flag instead of pretending that the logarithm removes $r = 0$.

Uniform log-radius steps become multiplicative real-radius shells

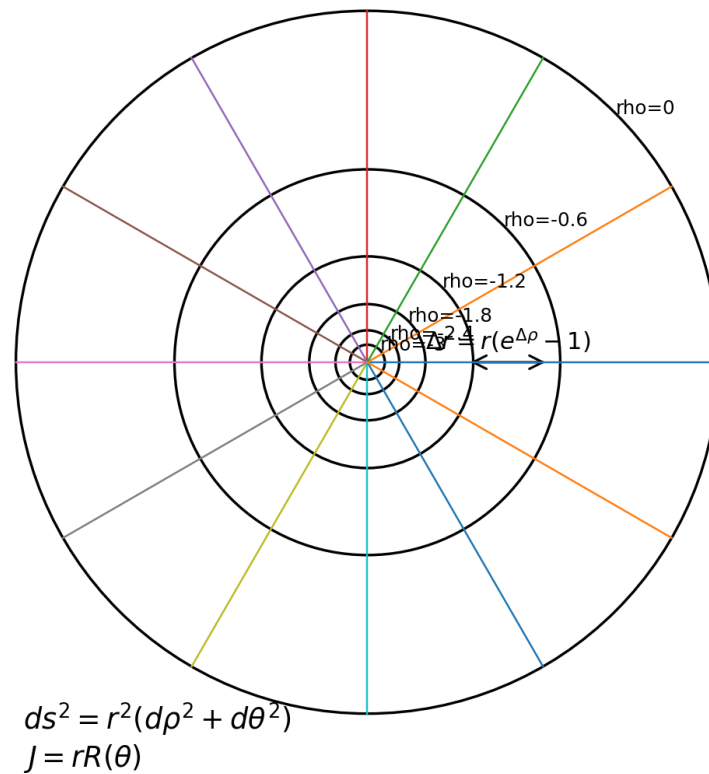


Figure 3: The log chart is a non-uniform coordinate representation, not infinite precision.

5.2 Metric and exact local step

The planar metric is conformal:

$$ds^2 = r^2(d\rho^2 + d\theta^2), \quad g(\rho) = r^2 I_2.$$

A finite log step gives the exact real-radius increment

$$\Delta r = r(e^{\Delta\rho} - 1),$$

not merely $r\Delta\rho$ except to first order.

5.3 Jacobian, velocity and change of change

The Jacobian is

$$J = r \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

With radial and angular unit vectors e_r, e_θ ,

$$v = r(\dot{\rho}e_r + \dot{\theta}e_\theta),$$

$$a = r[(\ddot{\rho} + \dot{\rho}^2 - \dot{\theta}^2)e_r + (\ddot{\theta} + 2\dot{\rho}\dot{\theta})e_\theta].$$

This is the exact normalized meaning of “change of change” in the chart. If a scalar field has chart derivatives (f_ρ, f_θ) , then

$$\nabla_{xy} f = \frac{f_\rho e_r + f_\theta e_\theta}{r}.$$

A coordinate component may scale as $e^{-\rho}$, but a physical force does not automatically do so: force requires a declared mechanics model and tensor transformation.

6 One-bit jitter and guard intervals

The source uses a one-bit analytical jitter as a high-frequency perturbation and as tree metadata. SCLP separates these roles. For seed, address K and tick X ,

$$b = H(\text{seed}, K, X) \bmod 2, \quad \sigma = 2b - 1.$$

An optional field perturbation is

$$f_j = f + \epsilon_j \sigma.$$

The authoritative relation is not discarded. Instead it receives the interval certificate

$$f \in [f_j - \epsilon_j, f_j + \epsilon_j] = [f - \epsilon_j, f + \epsilon_j].$$

The jitter profile is admissible only when $\epsilon_j < \epsilon_g$ and the interval cannot change the verified event class or ordering.

For the bundled example,

$$\begin{aligned} f &= -0.1700961894, \\ \epsilon_j &= 10^{-4}, \quad \epsilon_g = 10^{-3}, \\ [f - \epsilon_j, f + \epsilon_j] &= [-0.1701961894, -0.1699961894]. \end{aligned}$$

The entire interval remains negative, so the one-bit modulation cannot move this query across the zero relation.

7 Time thread, phase and winding

The source turns the 18:55 statement into a phase seed and a loop count. SCLP keeps the useful idea but not the claim that time itself stops being linear. Given reference tick X_0 , period P and linear tick X ,

$$u = \frac{X - X_0}{P}, \quad w = \lfloor u \rfloor, \quad \psi = u - w \in [0, 1).$$

Here ψ is the phase on $\mathbb{S}^1$ and w is the winding lineage. The reference profile declares 18:55 as 1,135 minutes from midnight; that unit is metadata, not a universal constant.

At $X = 1920$ and $P = 256$, the example produces

$$w = 3, \quad \psi = 0.06640625, \quad w \bmod 2 = 1.$$

Repeated phase coordinates therefore remain distinguishable by winding and lineage.

8 Topological wrapping: source form and Klein form

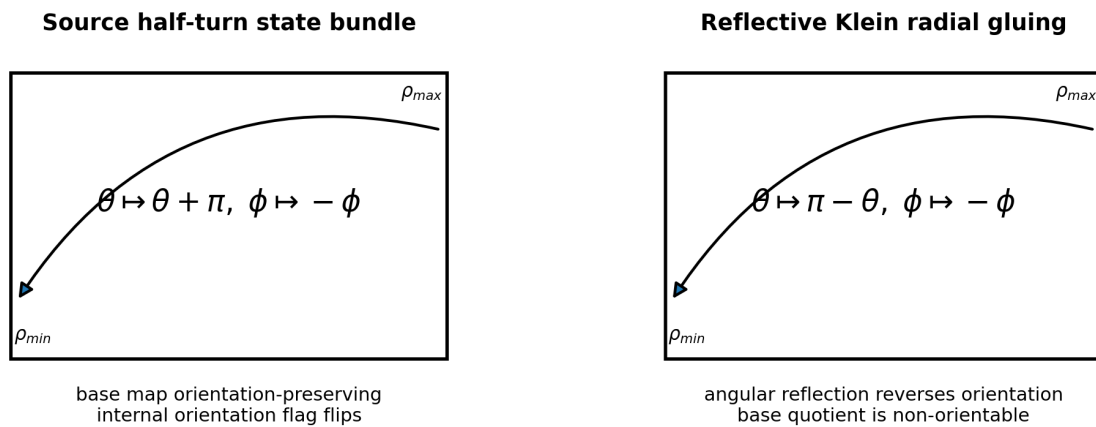


Figure 4: The source half-turn formula and the orientation-reversing quotient are represented as separate profiles.

8.1 Source half-turn bundle

On an odd radial wrap, the literal source formula is retained as

$$(\rho, \theta, \phi, o) \mapsto (\rho', \theta + \pi, -\phi, -o).$$

Because $\theta \mapsto \theta + \pi$ is a rotation of the angular circle, it does not by itself make the base coordinate quotient non-orientable. The internal hoop/orientation state does flip, so the operation is useful as a twisted state bundle.

8.2 Reflective Klein profile

The actual non-orientable profile uses

$$(\rho, \theta, \phi, o) \mapsto (\rho', \pi - \theta, -\phi, -o)$$

on odd radial wraps, with periodic angular closure on the other boundary. The angular reflection has derivative -1 , so the base gluing reverses orientation. The hoop coordinate and orientation flag flip with it.

For the shipped wrap example, $\rho = 0.25$ in the interval $[-20, 0)$ creates one wrap and returns

$$\rho' = -19.75, \quad \theta' = -\pi, \quad \phi' = -0.34906585, \quad o' = -1.$$

The chart transition changes coordinate components; it is not an instantaneous physical torque impulse.

9 Hinge calculus and the missing shackle

9.1 Typed hinge state

The hinge state is

$$h = (\phi, \omega, \alpha) = (\phi, \dot{\phi}, \ddot{\phi}).$$

Torque is not another name for angular acceleration. Only after a mechanical profile supplies inertia I , damping c and stiffness k may one compute, for example,

$$\tau = I\alpha + c\omega + k\phi.$$

The reflective chart map sends (ϕ, ω, α) to $(-\phi, -\omega, -\alpha)$ under the symmetric model.

9.2 Constraint release

Let a velocity constraint be

$$A(q)\dot{q} = 0.$$

The missing shackle is represented by removing one declared row to obtain A' . The exact freedom gain is

$$\Delta d = \text{nullity}(A') - \text{nullity}(A).$$

In the reference matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

removing the second row changes rank from 2 to 1 and nullity from 1 to 2, so $\Delta d = 1$. This establishes an additional legal velocity direction. It does not establish chaos, a force law or a floating-base trajectory without further dynamics.

9.3 Motion constrained to an SDF-zero coil

For a regular zero surface $f(x) = 0$ with unit normal n , a proposed velocity v can be kept tangent to first order by

$$P_T = I - nn^T, \quad v_T = P_T v,$$

which guarantees $n \cdot v_T = 0$. Numerical drift still requires projection or constrained integration.

10 Bit branch, bifurcation and bounded L-system

A branch guard returns $b \in \{0,1\}$ and selects a declared transition or production. A binary branch is not automatically a chaotic bifurcation. Chaos requires a specific dynamical map and evidence such as sensitivity or a positive Lyapunov exponent.

The reference parametric grammar rewrites each forward symbol $F(T)$ into two dyadically scaled branches with a signed turn, a balanced push/pop pair and a metric-aware jitter token. In schematic form,

$$F(T) \Rightarrow F(T/2) [\pm\phi F(T/2)] J(\Delta r).$$

The sign is selected by the branch bit and then multiplied by topological chirality. The automorphism $+\leftrightarrow-$ implements the source's turn reversal at the reflective boundary.

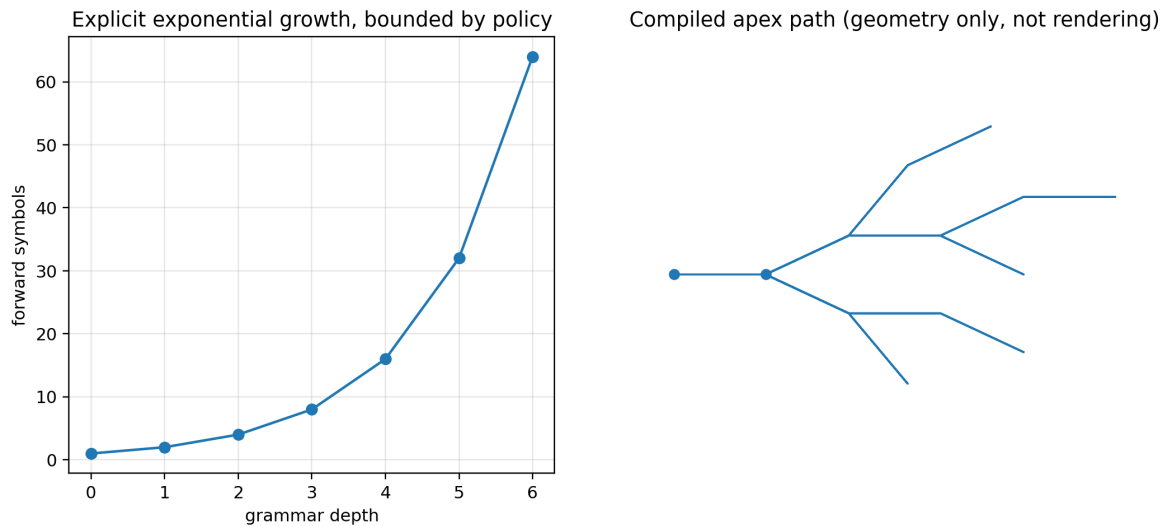


Figure 5: Grammar growth is explicit, finite and budgeted; the compiled line is an apex path, not a rendered scene.

The example depth is 4. It produces 16 forward symbols, 76 total symbols, maximum stack depth 4 and a 12-bit grammar-state word of 1058. The implementation rejects depth, symbol or stack overflow.

11 Bit shift as radix-trie refinement

The source says a left shift moves deeper into a 1-bit binary search tree and halves log-radius/angle. SCLP corrects the mechanism:

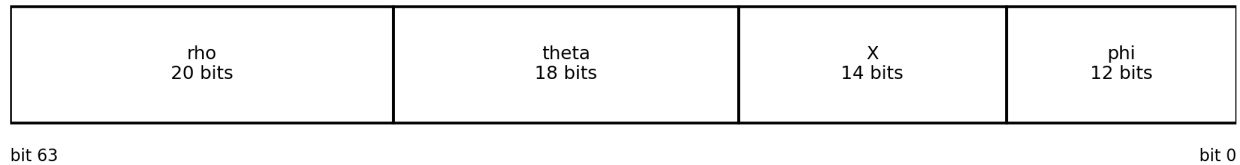
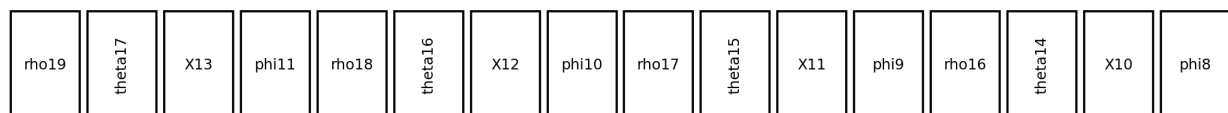
- the structure is a radix-2 trie over fixed-width key prefixes, not a comparison-based BST;
- appending a bit performs $p' = (p \ll 1) \mid b$;
- that bit fixes one scheduled coordinate bit in the Morton profile;
- the corresponding integer cell interval is refined; decoded physical coordinates are not directly divided by two.

A 64-level path is therefore a sequence of coordinate refinements. Prefix bounds are executable: after each appended bit, no field interval can widen, and at least the scheduled field narrows.

12 Two exact 64-bit key layouts

The source page 12 simultaneously supplies contiguous bit ranges and a Morton-style interleaving equation. Those are not the same layout. Version 3.6.2 keeps both under separate IDs.

Field	Bits	States	Contiguous range	Quantization
rho	20	1,048,576	63:44	$[-20,0]$; step $1.9073504518e-05$
theta	18	262,144	43:26	$[0,2\pi]$; step $2.39684498107e-05$
time	14	16,384	25:12	modular ticks; step 1
phi	12	4,096	11:0	$[0,2\pi]$; step 0.00153398078789

Table 1: Exact field widths for the 64-bit SCLP key.**Contiguous field layout: source bit-range table****Morton layout: first 16 output bits of the MSB round-robin schedule**

Two distinct encodings; both have tested pack/unpack round trips

Figure 6: The key layouts are distinct and have independent pack/unpack tests.

The contiguous word is

$$K_C = (q_\rho \ll 44) \mid (q_\theta \ll 26) \mid (q_X \ll 12) \mid q_\phi.$$

The Morton word consumes bits in MSB round-robin order:

$$\rho_{19}, \theta_{17}, X_{13}, \phi_{11}, \rho_{18}, \theta_{16}, X_{12}, \phi_{10}, \dots$$

until all 64 source bits are used. The complete schedule is included as CSV and JSON.

For the reference state,

$$\begin{aligned} (q_\rho, q_\theta, q_X, q_\phi) &= (949111, 0, 1920, 227), \\ K_C &= 0xe7b77000007800e3, \\ K_M &= 0x88823bb88099128b. \end{aligned}$$

Both decode exactly to the same quantized tuple, while the words are intentionally different.

EXACT METRICS

The finite address capacity is exactly

$$2^{20}2^{18}2^{14}2^{12} = 2^{64} = 18,446,744,073,709,551,616.$$

For $\rho \in [-20,0]$, the reference steps are $1.90735045 \times 10^{-5}$ in ρ , $2.39684498 \times 10^{-5}$ radians in θ , 16,384 modular time states and $1.53398079 \times 10^{-3}$ radians in ϕ .

13 Pointerless storage and compression audit

A dense complete binary heap can infer children $2i+1$ and $2i+2$ without pointers. A sparse tree cannot simply prune arbitrary gaps and keep that formula without either reserving the gaps or adding topology/presence information. A succinct production structure needs a topology bitvector, a payload bitvector and rank/select or equivalent navigation metadata.

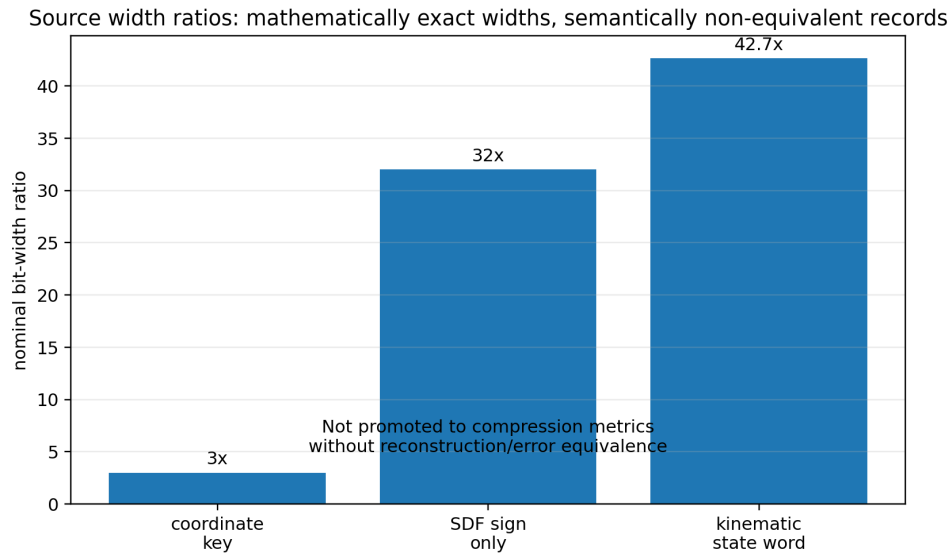


Figure 7: The source numbers are retained as width ratios and not promoted to equal-information compression.

The exact arithmetic is:

$$192/64 = 3, \quad 32/1 = 32, \quad 512/12 = 42.66\bar{6}.$$

But the compared records are not semantically equivalent:

- six float32 boundaries contain different information and error semantics from one quantized key;
- a one-bit sign does not retain SDF magnitude, exact zero state or uncertainty;
- a 12-bit state word can select only a finite versioned codebook, not encode an arbitrary 4×4 float matrix.

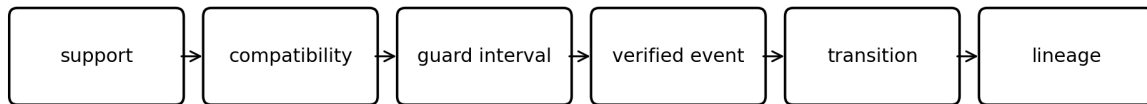
Eight raw 64-bit keys fit in a 64-byte cache line. This establishes capacity only, not eight evaluations in one cycle. The 64-bit profile is finite; it does not store infinite detail.

14 Literal referential substrate integration

The machine-readable example contains 22 content-addressed definitions. Each definition records ID, kind, domain, codomain, evaluation phase, dependencies, parameters, invariants, provenance and canonical SHA-256 content hash. Dependency ordering is acyclic and validated before execution.

The reference pipeline is:

1. select SCLP profile and typed symbol contract;
2. construct the finite cone and evaluate its exact relation;
3. test the paired-sphere support;
4. compute log-polar state, metric and phase/winding;
5. quantize and encode both 64-bit layouts;
6. refine a radix prefix and certify jitter;
7. evaluate source-twist and reflective-Klein profiles separately;
8. release the selected constraint row and evaluate hinge mechanics;
9. select the branch, expand the bounded grammar and certify the sweep;
10. audit width/memory claims;
11. issue an integrated certificate and hand off to UGTS.

SCLP 3.6.2 returns authority to the canonical UGTS query/event sequence

One-bit fields remain route/predicate metadata; continuous residual, uncertainty and identity remain separate

Figure 8: The new mechanisms do not replace the canonical event sequence.

The exact reference certificate is valid. Its notable fields are:

Field	Result
cone residual	-0.1700961894 (interior)
log-polar state	$\rho = -1.8971199849$, $\theta = 0$, metric scale 0.0225
jitter interval	$[-0.1701961894, -0.1699961894]$, safe under 0.001 margin
time	phase 0.06640625 , winding 3
Klein wrap	one wrap, orientation -1 , $\rho' = -19.75$
missing shackle	nullity $1 \rightarrow 2$, freedom gain 1
grammar	depth 4 , 16 forward symbols, 76 total, state word 1058
sweep	certified interval $[-0.3024764633, -0.2966170883]$
key layouts	exact round trip for both; distinct words

15 Reference implementation and verification

The package retains all 3.6 and course-corrected 3.6.1 files and adds:

- `src/ugts36/sclp362.py`: geometry, metric, topology, constraints, grammar, key and metric operators;
- `src/ugts36/sclp_runtime.py`: literal referential pipeline evaluator;
- `spec/SCLP_3_6_2_FORMAL_DEFINITION.md`;
- a 50-operator CSV/JSON catalog and 20-entry claims ledger;
- JSON Schema and content-addressed example substrate;
- exact key-layout and 64-row Morton schedule files;
- deterministic certificate, trace and demonstration;
- editable LaTeX source, figures, tests, validation report and SHA-256 manifest.

The 70 new tests cover exact cone geometry, paired spheres, sweep intervals, log-polar derivatives, jitter bounds, time/winding, both topology profiles, hinge mechanics, constraint rank/nullity, tangent projection, finite grammar budgets, both key codecs, prefix refinement, trie accounting, metric qualifications, JSON Schema, content hashes, pipeline ordering and requester attribution.

Together with the retained 83 tests, the combined package contains 153 passing tests. These establish internal consistency and executable bounds. They do not validate a physical device, a universal performance advantage or a claim of infinite detail.

16 Claims ledger and kill criteria

Table 2: SCLP 3.6.2 claims ledger.

ID	Source claim or motif	Disposition	Technical boundary
C362-01	A side length T alone defines the cone	CORRECTED	T is paired with a half-angle; $h=T \cos \alpha$ and $R=T \sin \alpha$.

ID	Source claim or motif	Disposition	Technical boundary
C362-02	The source theta+pi wrap alone is a Klein-bottle base quotient	BOUNDED	It is retained as an internal state-bundle half-turn. A reflective theta map is required for the non-orientable base profile.
C362-03	Time stops being linear and becomes a loop	CORRECTED	Linear time is retained; phase is time modulo a declared period and winding remains separate lineage.
C362-04	18:55 is an intrinsic physical constant	REJECT	It is a profile seed/reference tick with a declared unit.
C362-05	Bit shifting directly halves rho and theta	CORRECTED	Appending a trie bit refines one scheduled quantized interval; coordinate values are decoded separately.
C362-06	Every branch is chaotic	REJECT	A binary branch is deterministic. Chaos requires a specified map and measured sensitivity.
C362-07	A missing shackle automatically produces chaos	CORRECTED	Constraint-row deletion changes rank/nullity; dynamics require mass, force and damping models.
C362-08	All forces automatically scale by e^{ρ}	CORRECTED	Coordinate covectors transform by the inverse Jacobian; physical force laws remain separately typed.
C362-09	The sweep field is always an exact SDF	REJECT	The report provides an exact cone SDF and a certified interval for the sweep infimum; a generic sweep envelope is not automatically an exact SDF.
C362-10	One jitter bit contains roughness, sensor noise and full state	REJECT	The bit is a bounded deterministic perturbation or selector; residual, error, topology and lineage remain separate.
C362-11	The contiguous bit table is the same as Morton interleaving	CORRECTED	Two explicit layouts are supplied and tested independently.
C362-12	Pointerless sparse trees have zero metadata	REJECT	They avoid explicit child pointers but still require topology/presence and navigation metadata.
C362-13	192-to-64 is an equivalent 3:1 compression	BOUNDED	It is a nominal width ratio unless decode ranges, errors and semantic equivalence are declared.
C362-14	A sign bit is equivalent to a float SDF	REJECT	It discards magnitude and uncertainty and needs a separate boundary representation.
C362-15	A 12-bit word is equivalent to any 4x4 float matrix	REJECT	It can only select a versioned finite codebook/state family.
C362-16	Eight keys per cache line means eight outcomes per clock	REJECT	Capacity does not establish execution width, latency or instruction throughput.
C362-17	Log packing stores infinite detail in finite memory	REJECT	The declared profile contains exactly 2^{64} keys and finite grammar/state budgets.
C362-18	All possible movements are universal $O(1)$	REJECT	Only bounded direct lookup after key construction may be constant expected access; generation, collisions, verification and memory remain.
C362-19	L-system recursion is infinite in the substrate	CORRECTED	Expansion is finite and rejects depth, symbol or stack overflow.
C362-20	Rasterization/raymarching are required	EXCLUDED	The 3.6.2 core is purely referential geometry, topology, indexing and event calculus; display remains downstream.

The 3.6.2 delta is rejected or simplified for a target application if any of the following occurs:

- quantization or jitter exceeds the event guard margin or changes event ordering;
- the sweep interval remains too wide to classify the relation at the required tolerance;
- grammar growth or branch density exceeds the declared budget;
- key collisions, table occupancy or sparse-tree metadata erase the expected storage gain;
- the source half-turn profile is incorrectly treated as a Klein base quotient;
- the 12-bit grammar/state word cannot reconstruct the required state family;
- pointerless layout, decoding or verification costs exceed a conventional index at equal error;
- any width ratio is reported as compression without semantic and reconstruction equivalence;
- a physical torque, force, chaos or throughput claim is made without a specified model and measurement.

17 Final definition

FORMAL DEFINITION

UGTS-KC 3.6.2 SCLP is a finite, content-addressed, query-first delta in which a finite cone and paired spherical supports provide exact primitive relations; a translational sweep has a certified relation interval; a log-polar chart supplies a metric and kinematic calculus; one-bit jitter is bounded by an explicit guard interval; linear time maps to phase plus winding; topological wrapping is split into a literal half-turn bundle and a reflective Klein quotient; hinge and missing-shackle behavior are typed by angular state and constraint rank; branch bits drive a finite budgeted grammar; and 20/18/14/12 quantized fields are encoded by two explicit, invertible 64-bit layouts. Every result is passed to support, compatibility, guard, event, transition and lineage rather than to a renderer.

EVIDENCE BOUNDARY

This document does not assert that the source's entire combined apparatus is a previously established physical theory. It provides a clean formal profile in which the recoverable geometric, topological, kinematic and packing mechanisms are explicit, testable and falsifiable.

A Complete SCLP 3.6.2 operator catalog

Table 3: Complete 50-operator SCLP 3.6.2 delta catalog.

Operator ID	Domain	Mechanism	Normalized technical definition	Disposition	Source
sclp362.profile.packed-swept-cone-v1	Profile	Packed SCLP profile	Typed finite profile combining a cone relation, log-polar chart, modular time, topology, grammar and two 64-bit key layouts.	ENGINEERING-DERIVED	pp.9-13
sclp362.type.symbol-separation-v1	Typing	Overloaded-symbol separation	T is slant length; t is linear time; X is a modular tick; phi is a hoop angle; delta_rho and epsilon_guard are distinct.	CORRECTED	pp.1-8
sclp362.type.lowercase-phi-circle-v1	Typing	Lower-case phi as S1 coordinate	Represent phi as a periodic angle in $R/(2\pi Z)$; do not identify it with the golden ratio Phi.	RETAIN	pp.1,3,7-10
sclp362.type.delta-family-v1	Typing	Typed delta family	Separate log-radius increment, time increment, finite-difference step, jitter amplitude and guard margin.	CORRECTED	pp.1-3,7-8
sclp362.type.evidence-boundary-v1	Governance	Source/normalization boundary	Every claim is labeled source-derived, corrected, bounded or rejected and carries a profile scope.	ENGINEERING-DERIVED	all
sclp362.geometry.cone-slant-angle-v1	Geometry	Cone from slant length and half-angle	Given $T > 0$ and alpha in $(0, \pi/2)$, derive height $h = T \cos \alpha$ and base radius $R = T \sin \alpha$.	CORRECTED	p.1
sclp362.geometry.finite-cone-sdf-v1	Geometry	Exact finite-cone signed distance	Reduce to signed distance from the meridian point to the filled triangle $(-R, h), (R, h), (0, 0)$.	ENGINEERING-DERIVED	pp.1,8,10
sclp362.geometry.cone-relation-class-v1	Geometry	Three-state relation class	Classify a finite-cone residual as interior, guard-zero or exterior with a declared epsilon.	RETAIN	pp.9-10
sclp362.geometry.sphere-sdf-v1	Geometry	Sphere relation	Use $ x - c - R$ as an exact local spherical relation.	RETAIN	p.1
sclp362.geometry.paired-sphere-support-v1	Support	Two-sphere support pair	Treat the two-sphere eye motif as two local spherical supports with union/intersection and overlap tags.	TRANSLATE	p.1
sclp362.sweep.translation-family-v1	Sweep	Rigid translational cone family	$C(u) = C_0 + s(u)$ with fixed orientation and a declared finite interval.	TRANSLATE	pp.1,6-8
sclp362.sweep.lipschitz-interval-v1	Sweep	Certified sweep interval	Uniform samples plus the 1-Lipschitz translation law bound the true sweep infimum in $[m - Lh/2, m]$.	ENGINEERING-DERIVED	pp.1,6-8
sclp362.sweep.zero-tangent-projector-v1	Kinematics	Zero-surface tangent projection	Project a proposed velocity with $P = I - n n^T$ so its first-order field derivative vanishes on a regular zero set.	TRANSLATE	pp.3-4
sclp362.logpolar.chart-v1	Coordinates	Log-polar chart	$\rho = \ln(r/r_0)$, $\theta = \text{atan2}(y, x)$ with an explicit core radius.	RETAIN	pp.2-3,8-10
sclp362.logpolar.core-v1	Coordinates	Explicit origin core	The logarithmic chart does not remove the $r=0$ singularity; a core flag and radius are mandatory.	CORRECTED	p.2
sclp362.logpolar.metric-v1	Metric	Conformal metric tensor	In the plane $ds^2 = r^2(d\rho^2 + d\theta^2)$, so the metric factor is $r^2 = e^{2\rho} r_0^2$.	ENGINEERING-DERIVED	pp.2,9-10
sclp362.logpolar.exact-radial-step-v1	Metric	Exact local radial increment	A log step $\Delta \rho$ gives $\Delta r = r(\exp(\Delta \rho) - 1)$, approximately $r \Delta \rho$ only to first order.	CORRECTED	p.2
sclp362.logpolar.jacobian-v1	Kinematics	Log-polar Jacobian	$J = r[\cos \theta, \sin \theta], [\sin \theta, \cos \theta]$.	ENGINEERING-DERIVED	pp.3,8,10
sclp362.logpolar.velocity-v1	Kinematics	Velocity transform	$v = r(\rho_{\dot{}} e_r + \theta_{\dot{}} e_{\theta})$.	ENGINEERING-DERIVED	p.3
sclp362.logpolar.acceleration-v1	Kinematics	Change-of-change transform	$a = r[(\rho_{\ddot{}} + \rho_{\dot{}}^2 - \theta_{\dot{}}^2) e_r + (\theta_{\ddot{}} + 2 \rho_{\dot{}} \theta_{\dot{}}) e_{\theta}]$.	ENGINEERING-DERIVED	pp.1,3,8,10
sclp362.logpolar.gradient-v1	Metric	Gradient covector transform	Cartesian gradient equals $(f_{\rho} e_r + f_{\theta} e_{\theta})/r$; physical force scaling still requires a model.	CORRECTED	p.10
sclp362.jitter.deterministic-bit-v1	Topology/control	Deterministic one-bit jitter	Hash address, time tick and seed to a reproducible bit used as a bounded signed perturbation or selector.	TRANSLATE	pp.1-6
sclp362.jitter.interval-guard-v1	Event calculus	Jitter interval certificate	Authoritative residual f is enclosed by $[f - \epsilon_j, f + \epsilon_j]$; ϵ_j must remain below the event margin.	CORRECTED	pp.1-3
sclp362.time.phase-clock-v1	Time	Linear-time phase clock	Keep linear time and map it to phase plus integer winding; the source timestamp is profile metadata with a declared unit.	CORRECTED	pp.3-5
sclp362.time.winding-lineage-v1	Persistence	Winding lineage	Record winding count and parity separately from periodic phase so repeated coordinates retain history.	TRANSLATE	pp.3-4
sclp362.topology.source-half-turn-v0	Topology	Source half-turn bundle twist	On odd radial wraps apply $\theta + \pi$, $\phi \rightarrow -\phi$ and orientation flip; base gluing is not itself a Klein quotient.	RETAIN/BOUND	pp.9-10
sclp362.topology.klein-reflection-v1	Topology	Reflective Klein radial gluing	Use $\theta \rightarrow \pi - \theta$ plus ϕ and orientation reversal on odd radial wraps to obtain an orientation-reversing quotient.	CORRECTED	pp.2-4,9-10
sclp362.topology.tangent-chirality-v1	Topology	Tangent chirality map	The chart transition sends $d\theta$ and $d\phi$ to their negatives on odd reflective wraps; it is not a physical impulse.	ENGINEERING-DERIVED	pp.4,6,10

Operator ID	Domain	Mechanism	Normalized technical definition	Disposition	Source
sclp362.topology.wrap-event-v1	Event calculus	Topological wrap event	Crossing rho_max emits a transition with coordinate map, orientation patch, wrap count and lineage.	TRANSLATE	pp.2-5,9-11
sclp362.hinge.state-v1	Kinematics	Hinge state phi, omega, alpha	Store angle, angular velocity and angular acceleration as separate typed coordinates.	CORRECTED	pp.1,3,8,10
sclp362.hinge.torque-model-v1	Kinematics	Optional physical torque model	Only after declaring inertia, damping and stiffness may torque be computed as $I \alpha + c \omega + k \phi$.	CORRECTED	pp.1,3,8
sclp362.constraint.row-release-v1	Kinematics	Missing-shackle row deletion	Model the missing shackle as removal of one constraint-Jacobian row.	TRANSLATE	pp.1,3-6,8-10
sclp362.constraint.nullity-gain-v1	Kinematics	Freedom-gain certificate	Compare nullities before and after row deletion; one independent row can add one degree of freedom.	ENGINEERING-DERIVED	pp.1,3,8,10
sclp362.branch.binary-guard-v1	Event calculus	Binary guard branch	A declared guard partitions state into branch 0 or 1 and selects a transition/grammar rule.	TRANSLATE	pp.4-7
sclp362.branch.no-chaos-inference-v1	Audit	No automatic chaos inference	Binary branching is deterministic; chaos requires a defined dynamical map and measured sensitivity/Lyapunov evidence.	CORRECTED	p.5
sclp362.grammar.bounded-binary-lsystem-v1	Grammar	Bounded binary parametric system	Each F emits two dyadically scaled branches, a typed turn and jitter token under depth, symbol and stack budgets.	TRANSLATE	pp.6-8
sclp362.grammar.chirality-automorphism-v1	Grammar	Chirality turn automorphism	A topological orientation flip maps positive turns to negative turns and vice versa.	RETAIN	pp.6-7
sclp362.packing.quantize-20-18-14-12-v1	Packing	20/18/14/12 quantizer	Quantize rho, theta, modular time and phi into exactly 64 field bits with declared ranges and errors.	RETAIN/BOUND	pp.11-13
sclp362.packing.contiguous-key-v1	Packing	Contiguous field key	Pack fields into source table ranges [63:44],[43:26],[25:12],[11:0].	RETAIN	p.12
sclp362.packing.morton-key-v1	Packing	MSB round-robin Morton key	Interleave rho19,theta17,time13,phi11, then continue round-robin until all 64 bits are consumed.	RETAIN/CLARIFY	pp.10-12
sclp362.packing.layout-separation-v1	Packing	Key-layout separation	The contiguous table and interleaved Morton word are two distinct encodings, never silently treated as the same bit layout.	CORRECTED	pp.10-12
sclp362.index.prefix-refinement-v1	Indexing	Radix prefix refinement	Appending a bit fixes one scheduled coordinate bit and refines a quantized cell; it does not directly divide rho or theta values.	CORRECTED	pp.4-5
sclp362.index.radix-trie-v1	Indexing	Radix-2 trie terminology	Use a radix trie over key prefixes rather than calling the structure a comparison-based binary search tree.	CORRECTED	pp.1-12
sclp362.index.one-bit-payload-v1	Indexing	Narrow one-bit payload	A leaf bit may encode route, occupancy class or predicate result; continuous residual, uncertainty and lineage remain external.	CORRECTED	pp.1-2,9-10
sclp362.index.sparse-presence-accounting-v1	Memory	Sparse presence accounting	Pointerless sparse storage still needs topology/presence bits and rank/select or equivalent navigation metadata.	CORRECTED	pp.11-12
sclp362.metrics.cacheline-eight-keys-v1	Metrics	Eight raw keys per cache line	A 64-byte line stores eight 64-bit keys; this does not imply eight evaluations in one cycle.	CORRECTED	p.13
sclp362.metrics.nominal-width-audit-v1	Metrics	Nominal width audit	Report 3x,32x and 42.666x only as bit-width ratios because the compared records do not preserve equal semantics.	CORRECTED	pp.12-13
sclp362.metrics.finite-capacity-v1	Metrics	Finite 2^{64} key capacity	The 20+18+14+12 profile has exactly 2^{64} address combinations, not infinite detail.	CORRECTED	pp.12-13
sclp362.query.direct-lookup-bound-v1	Query	Bounded direct lookup	Fixed-width array/hash lookup may be constant expected RAM access after key construction; traversal, decoding, verification and storage remain costs.	BOUNDED	pp.7-11
sclp362.query.ugts-handoff-v1	Architecture	Canonical UGTS handoff	SCLP outputs feed support, compatibility, guard, verified event, transition and lineage in that order.	ENGINEERING-DERIVED	p.9-11

B Package reproduction

From the package root:

```
PYTHONPATH=src python tools/build_sclp362_assets.py
PYTHONPATH=src python tools/generate_sclp362_figures.py
python tools/generate_sclp362_tex_tables.py
PYTHONPATH=src python -m unittest discover -s tests -v
PYTHONPATH=src python examples/demo_sclp362.py
```

The PDF is built from report/UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.tex. The raw uploaded source is represented by hash and page-level notes and is not copied into the ZIP.